

Lab 7 Final Design Project

Reference Resource (optional and available in the GWC lab):

Practical Electronics for Inventors, Paul Scherz and Simon Monk, 2016, ISBN 978-1-25-958754-2

Design Project Overview:

Sparky Solar designs solar panel systems and requires a portable power meter that automatically measures and displays solar panel voltage, current, and power using an LCD. The goal is to improve usability, reduce calculation errors, and demonstrate innovative design.

Your team has been hired to design and build this automated solar panel power meter. The prototype will test a single solar panel with a maximum output of approximately 6 V. All testing and simulation will be conducted in Goldwater Center, room 185, using provided components. Final evaluation will be based on technical correctness, demonstrated understanding, and value added for stakeholders. Teams will demonstrate their working prototype to a Sparky Solar representative (TA) during weeks 14–16.

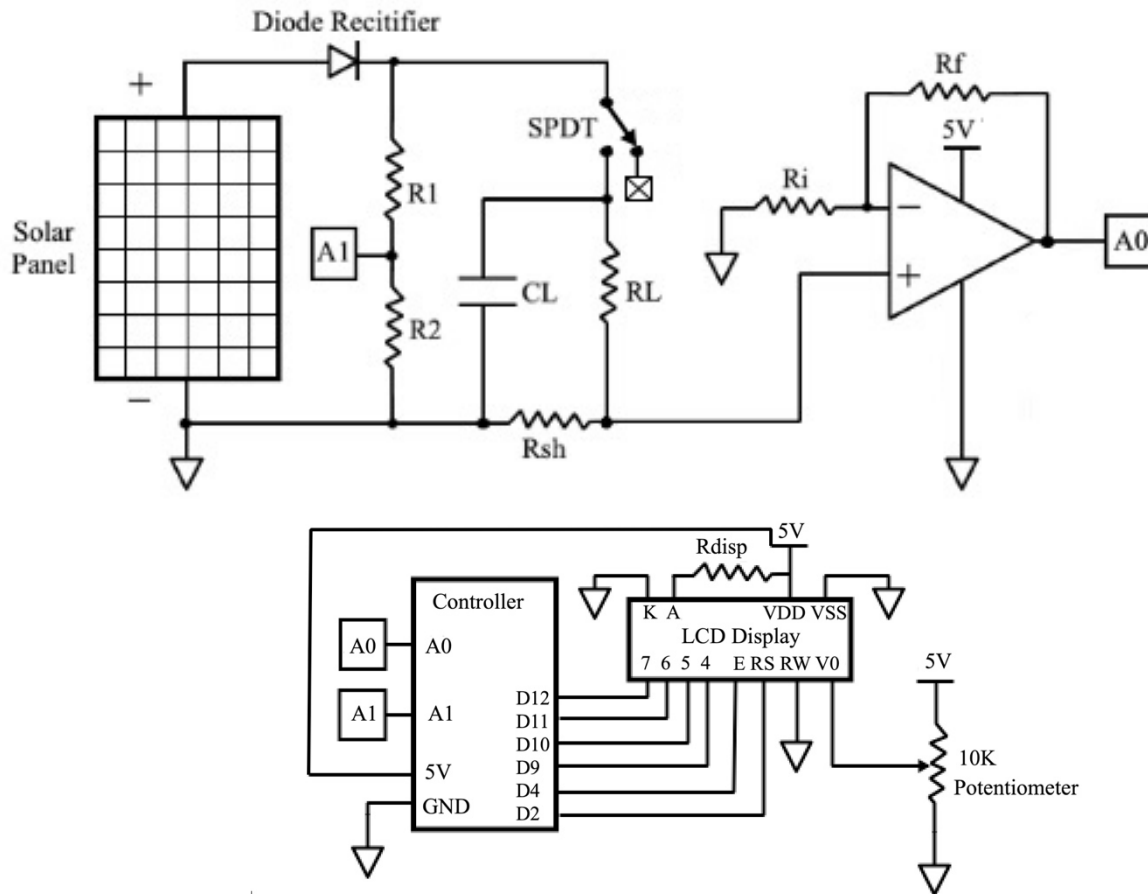
Inventory List:

The following is a list of available inventory for this project:

Part	Details
Resistors	10 Ω (10W), 47 Ω , 100 Ω , 220 Ω , 330 Ω , 510 Ω , 1k Ω , 1.5k Ω , 2.2k Ω , 3.3k Ω , 4.7k Ω , 10k Ω , 22k Ω , 47k Ω , 100k Ω , 220k Ω , 470k Ω , 1M Ω , 2M Ω
Capacitors	0.1 μ F, 1 μ F, 10 μ F, 22 μ F, 47 μ F, 100 μ F, 220 μ F
Diode Rectifiers	1N4007 (1N4148 in TinkerCAD and LTSPICE)
Op-Amp	Microchip MCP6022 or Analog Devices OP484
Alligator clips	Connectors
Assorted jumper wire	Red, black, white, green, blue, orange
Potentiometer	10K
Slide Switch	Breadboard-friendly SPDT Slide Switch
Solar Panels	6V, 167mA Solar Panel
DC Supply	LPS 161A DC Supply
Breadboard	830 Tie Points
Digital Multimeter	HP or Agilent 34401A (in lab) or Portable Multimeter
Oscilloscope	Tektronix MDO3012
UNO Controller Board	Elegoo or Arduino
USB Cable	Controller board to USB
LCD Display	LCD1602 Module with Pin Header
LEDS, buttons, buzzers, photoresistors	Assorted

Lab Instructions:

Sparky Solar has provided a basic schematic for the design as shown below:

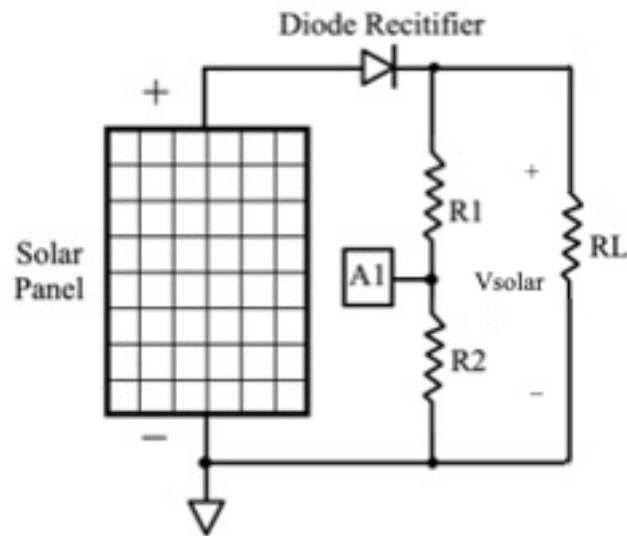


As the design team, you may modify the given design, provided all Sparky Solar requirements are met. This solar panel power meter must be designed to do the following:

1. Step down the voltage from the solar panel to a voltage value that can be an input into the controller (A1 = 5V maximum) for measurement. (*Lab 2 Voltage Dividers*)
2. Sample of the load current through a shunt resistor (R_{sh}), measure the voltage across the shunt resistor using the non-inverting Op-Amp, multiply the value of that voltage to a value that can be input into the controller ($0V < \text{Op-Amp output} < 5V \rightarrow \text{Op-Amp Gain} \sim 5\text{-}10V/V$), and input into the controller (A0). (*Lab 2 Resistor Networks, Lab 4 Superposition/Thevenin/Norton, Lab 5 Op-Amps*)
3. Use the Controller to measure and calculate the voltage, current, and power supplied to the load resistor (R_L) from the solar panel. (*Lab 3 Arduino*)
4. Output the voltage, current, and power values to the LCD display. (*Lab 3 Arduino*)
5. Select a load capacitor (C_L) so that the discharge time of the load (with a supply of 6V and 10mA) is about 0.01 second. The single pole dual throw switch (SPDT) is used to disconnect the load for discharging. (*Lab 6 Transient Circuits*)

All resistors selected for this design will be single resistors, not resistor networks. The following instructions are the detailed procedures of the summarized five steps above.

Step 1: Design the Voltage Divider



Design and test a **voltage divider** that scales the solar panel voltage to a maximum of 5V at Controller input A1 for all light conditions. You may test using a DC supply, a solar panel with a high-power lamp, or outdoor sunlight. Solar panels are limited; coordinate with your TA and record the panel number used (so you always test with the same panel). Verify the resistor power ratings as in Lab 2.

Design Recipe: Determine the Available Voltage After the Diode

Goal: Include the diode forward-voltage loss when calculating the available voltage for the voltage divider, shunt resistor, and op-amp stages.

1. Determine the Diode Voltage Drop

Diode Type:

Diode Type = _____

From the diode datasheet online, note the forward voltage at the expected current:

$V_D \approx$ _____ V

Typical Values:

- Silicon diode: $V_D \approx 0.6 - 0.7V$
- Schottky diode: $V_D \approx 0.2 - 0.4V$
- Rectifier: $V_D \approx 0.6 - 1V$

2. Determine the Available Voltage After the Diode

Measure the maximum solar panel voltage:

$V_{panel_max} =$ _____ V

Subtract the diode drop:

$V_{solar} = V_{panel_max} - V_D$

Data Sheet Part 1

Answer all the questions in the data sheet.

a) A diode is a device that allows current to only flow in one direction (*be sure to install the diode in the correct direction*). **What do you think the purpose of the diode rectifier is in this design?** (*In TinkerCAD or LTSpice, you can use the 1N4148 diode if the 1N4007 is not available*). **What voltage drop are you assuming across the diode (V_D)?**

b) Connect the load resistor ($R_L = 1\text{ k}\Omega$) to the solar panel and **measure the voltage and current** under different light conditions. Measure the voltage in parallel and the current in series (see Lab 2). Start current measurements on the highest range to avoid blowing the multimeter fuse. Use the portable DMM for outdoor measurements and the lab multimeter indoors. **Record all measurements on the data sheet and note the solar panel number used; use the same panel for all tests.**

c) The CEO of Sparky Solar, Melanie Waters, needs the voltage divider designed soon, so they can start testing. The footprint on the printed circuit board is tight, so the resistor must be small. There are no specific temperature requirements for this part of the design, but lower cost parts would be preferred. She has told you that the design criteria for the voltage divider (in order of importance) are:

1. Sooner availability, 2. Lower cost, 3. Smaller physical size.

When designing the voltage divider circuit, select which group of resistors you will use:

Resistors	Cost	Physical Size	Max Operating Temp	Availability
Set 1 (1Ω - 100Ω)	\$1.00/each	3mm x 1 mm	155°C	2-week delay
Set 2 ($1\text{k}\Omega$ - $100\text{k}\Omega$)	\$1.00/each	4mm x 3mm	155°C	Available now
Set 3 ($200\text{k}\Omega$ - $10\text{M}\Omega$)	\$1.50/each	2mm x 2mm	135°C	Available now

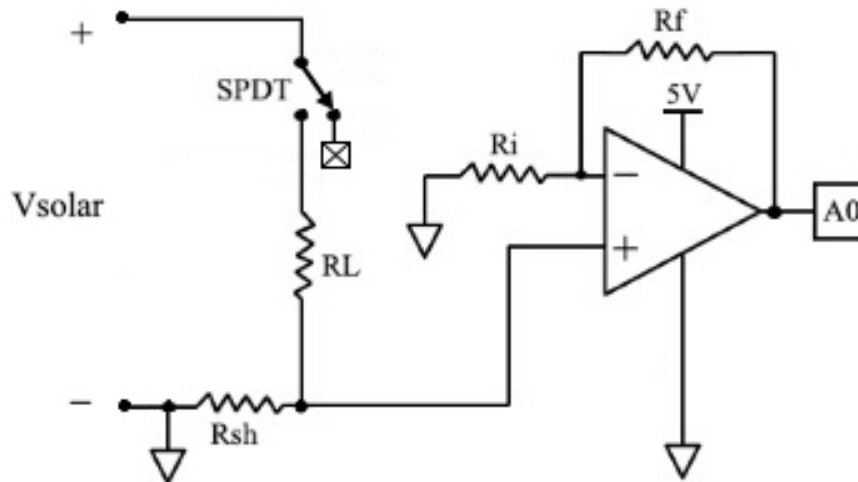
d) From your selected resistor set and inventory, **select R1 and R2** to ensure that the value of A1 does not exceed 5V for all light conditions. Using your maximum solar panel voltage, **what is the voltage into pin A1?** Which light condition should be considered when determining the values of R1 and R2 and why?

e) Design the resistor divider so the **maximum voltage** at pin A1 does not exceed **5V**. Verify that the maximum voltage across each resistor results in power dissipation below the **$\frac{1}{4}\text{ W}$ rating**. If either limit is exceeded, reselect R1 and R2. This check ensures the design operates within component limits.

f) Create a **simulation of your circuit in LTSPICE or TinkerCAD** and include a **screenshot with the data sheet**. You can replicate the Solar Panel with a DC supply in simulation. *Be sure to set your supply current high enough in the simulation to maintain the solar panel voltage.*

g) **Physically build and test your circuit.** You can use a DC supply or solar panel and light source in the lab to replicate the solar panel outside for troubleshooting and testing your design. Do your measured results from your built circuit match your simulated results? How much error are you seeing? What are some possible reasons for the difference in the real and theoretical results?

Step 2: Design the Op-Amp Circuit



Using your understanding of **Op-Amp circuits**, design and test a circuit that will measure the current across the load resistor (R_L). For this design, you can make the following assumptions:

- The op-amp is ideal ($V^+ = V^-$ and $I^+ = I^- = 0A$)
- The load resistance (R_L) is **1 k Ω**
- This circuit is in steady state (DC supply) so for your design you can assume the capacitor looks like an open for this part of the design
- To convert the load current into an input voltage for the Op-Amp, a shunt resistor (R_{sh}) of **50 Ω** is used
- The output voltage of the Op-Amp will be the input for the Controller calculation (A0)
- Read and understand the data sheet and pin diagram for the Op-Amp you are using (see Lab 5)

Data Sheet Part 2

Answer all the questions in the data sheet.

a) A small shunt resistor converts load current and convert it into a voltage, which is amplified by an op-amp and sent to the Arduino. for the op-amp. The shunt resistor must be very small so that almost no energy intended for the load resistor is wasted in measuring. Explain why an **op-amp** is required for this current measurement rather than how the voltage was measured in Part 1? (*Think about the reasons why op-amps are used*).

b) Determine the **maximum current** across your load resistor (R_L). The resistors provided are $\frac{1}{4}$ W resistors – will your resistors exceed this power rating (*compare with the currents from the solar panels in Part 1*)? (*The shunt resistor is small enough that you can omit it from your calculation*).

c) Using the highest load current from Part b, determine the **voltage into the + terminal** of the op-amp (*note: $V^+ = I_{load}R_{sh}$*). *The shunt resistor is very small because you don't want to draw extra current from the solar panel supply to do the measurements. However, this means that the op-amp input voltage will also be very small.*

d) The op-amp used for this design operates rail-to-rail on a 5V supply, so V_{out} must remain between 0 – 5V (anything outside that range will be “cut-off”). When selecting the feedback resistors, CEO Melanie Waters, has specified the circuit needs to function at temperatures as high as 155°C, should be available soon for testing, and a lower price point, if possible. The specifications for the circuit (in order of importance) are: **1. Higher operating temp, 2. Sooner availability, 3. Lower cost.**

When designing the op-amp circuit, select which group of resistors you will use for R_i and R_f to amplify the magnitude of the op-amp output voltage (*note: this is a non-inverting configuration*).

Resistors	Cost	Physical Size	Max Operating Temp	Availability
Set 1 (1Ω-100Ω)	\$1.00/each	3mm x 1 mm	155°C	2-week delay
Set 2 (1kΩ - 100kΩ)	\$2.00/each	4mm x 3mm	155°C	Available now
Set 3 (200kΩ - 10MΩ)	\$2.00/each	2mm x 2mm	135°C	Available now

e) As noted above, V_+ will be small, so you need to increase the output voltage of the op-amp so that the possible output voltages for all light conditions are better distributed in the 0-5V range. You still need to ensure that the output value of the op-amp does not exceed its supply voltage (5V) for the highest possible V_+ . Select **values for R_i and R_f that ensure the output voltage of the op-amp doesn't exceed the 5V limit** of the Arduino pin.

f) Why is a **rail-to-rail Op-Amp** required for this design? A rail-to-rail op-amp costs more than a standard op-amp. What value to Sparky Solar does this extra feature add to the design? Refer to *Lab 5: Op-Amps* to recall what a rail-to-rail op-amp is and why we use them. (*Hint: the voltage we sample from the shunt resistor is very small*).

g) Do the values of R_i and R_f need to be extremely accurate for your **current measurement to be precise**? Explain why or why not.

h) **Create a simulation of your circuit in LTSPICE or TinkerCAD** and include a screenshot with the data sheet. *Be sure to set you supply current in your simulation high enough to maintain the solar panel voltage.*

Before proceeding, complete the measurement setup checklist below. Failure to follow these steps may result in incorrect data or damaged equipment.

- Verify **common ground** between power supply, Arduino, and measurement instruments
- Confirm multimeter mode **before connecting** (V vs A)
- For current measurements:
 - Start on the **highest current range**
 - Break the circuit and insert the meter **in series**
- For voltage measurements, measure **in parallel** with the component
- Power **OFF** before moving meter leads or changing wiring
- Confirm expected current is **below meter and component ratings**

i) **Physically build and test your circuit.** You can use a DC supply to replicate the solar panel for troubleshooting and testing your design. Do your measured results from your built circuit match your simulated results? How much error are you seeing? What are some possible reasons for the difference in the real and theoretical results?

Design Recipe: Op-Amp Gain for Shunt Current Measurement

Goal: Select the op-amp gain so the small voltage across the shunt resistor maps safely into the Arduino's 0 – 5V analog input range.

1. Find the Maximum Shunt Voltage

Estimate the maximum load current:

$$I_{Lmax} = \text{_____} mA$$

Choose a shunt resistor:

$$R_{sh} = \text{_____} \Omega$$

Compute:

$$V_{shmax} = I_{Lmax} * R_{sh}$$

2. Determine the Required Gain

The Arduino input must satisfy:

$$V_{out} \leq 5V$$

Choose the gain:

$$G \approx \frac{5V}{V_{shmax}}$$

3. Select the Op-Amp Resistors

For a non-inverting amplifier:

$$G = 1 + \frac{R_f}{R_i}$$

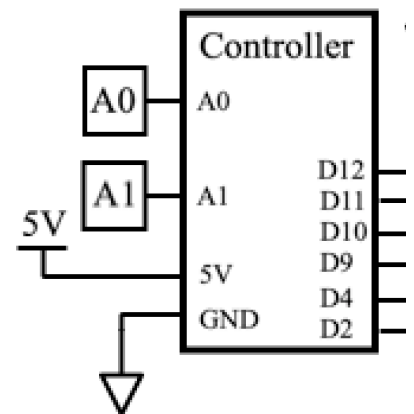
4. Verify Before Building

$$V_{out_max} = G * V_{shmax} < 5V$$

Step 3: Design the Controller Circuit

Use separate supplies for the solar panel/DC source and the Arduino. Never power the Arduino from the battery and USB simultaneously. **Share grounds, but do not mix supply voltages.**

Design and test an **Arduino circuit** with two analog inputs (A0 for current and A1 for voltage), a 5V output supply to power the op-amp, and code that displays current, voltage, and power on the Serial Monitor. Use the provided starter code and update it with **your selected resistor values**. The Lab 7 Arduino sketch (.ino) is provided [here](#). *You are encouraged to modify the code to better meet the design objectives.*



Additional Information regarding the Controller:

- The Controller is supplied with energy (voltage, current, power) from either the USB port or the 9V battery port, **not the 5V pin**
- The 5V pin is an **output pin** that can be used to supply other circuits (low current supply)
- Be sure to connect the Controller ground to your other grounds in the circuit to avoid floating nodes (*if you get odd voltage values, check your grounds*)
- Pins A0 and A1 are used as **analog inputs** and accept voltages from 0–5 V. The Arduino converts these analog values to digital values using a 10-bit A/D converter (1024 levels). In code, this conversion uses a factor of 5/1024. For example, a 1 V input corresponds to approximately 205 counts.

```

float Rf = ; //enter value of feedback resistor in Ohms
float Ri = ; //enter value of bias resistor in Ohms
float R1 = ; //enter value of series resistor in Ohms
float R2 = ; //enter value of load resistor in Ohms
float Rsh = 50; //enter value of shunt resistor in Ohms
float Read_Vtemp = A1; //value read from ADC on pin A1
float Read_Itemp = A0; //value read from ADC on pin A0
float Gain; //Gain of the Op-Amp Circuit
float Vtemp; //setting temporary analog voltage input
float Itemp; //setting temporary analog current input (actually a voltage reading)
float Voltage = 0.0; //initializing voltage reading
float Current = 0.0; //initializing current reading
float Power = 0.0; //initializing power calculation

void setup()
{
    Serial.begin(9600); //set up serial monitor
}

void loop()
{
    Vtemp = analogRead(Read_Vtemp);
    Itemp = analogRead(Read_Itemp);

    // Voltage Measurement
    // 1. Converts the analog input (max 5V) into a digital value (max 1024)
    // 2. Accounts for the voltage divider calculation
    Voltage = Vtemp * (5.0/1024.0) * (R1+R2) / R2; //in Volts

    // Current Measurement
    // 1. Calculates the Op-Amp Gain
    Gain = 1+(Rf/Ri); //in V/V
    // 2. Converts the analog input (5V max) into a digital value (max 1024)
    // 3. Accounts for the op-amp gain, current to voltage conversion (Rsh), and units (mA=1/1000)
    Current = Itemp*(5.0/1024.0)/(Gain*Rsh)*1000; //in mA

    // Power Measurement
    Power = Voltage * Current; //Calculation of power from solar panel in mW

    // Print values to the serial monitor for verification
    Serial.print("ADC V reads: ");
    Serial.print(Vtemp);
    Serial.print("    Solar Panel Voltage is: ");
    Serial.print(Voltage);
    Serial.print(" Volts");
    Serial.println("");
    Serial.println(" ");
    delay(1000);

    Serial.print("ADC I reads: ");
    Serial.print(Itemp);
    Serial.print("    Solar Panel Current is: ");
    Serial.print(Current);
    Serial.print(" milliAmps");
    Serial.println("");
    Serial.println(" ");
    delay(1000);

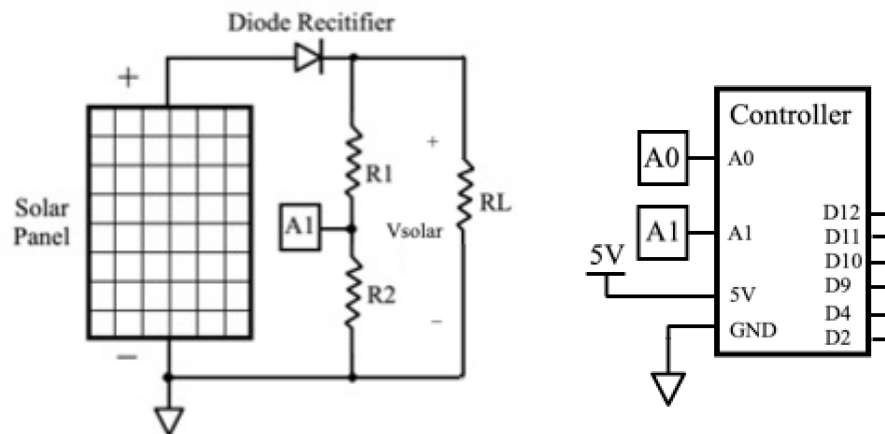
    Serial.print("Solar Panel Power is: ");
    Serial.print(Power);
    Serial.print(" milliWatts");
    Serial.println(" ");
    Serial.println("_____");
    delay(1000);
}

```


Data Sheet Part 3

Answer all the questions in the data sheet.

- Explain the **Voltage and Current calculations** in the Arduino code provided in your own words.
- Record the **resistor values** you are using for your voltage divider and op-amp feedback designs. (from Parts 1 and 2)
- Using the voltage, current, and power calculations from the Arduino code provided, **calculate the expected Serial Port voltage, current, and power values** for the given A0 and A1 inputs. (What will the Arduino output for "Voltage", "Current" and "Power"?). These values will depend on your R_1 , R_2 , R_i , and R_f design choices. Hint: the equations you need are in the Arduino Code that you've been provided – but don't include the analog to digital conversion ($5/1024$) as this is used to convert the 5V input into 1024 digital bits)
- Create a **simulation of your Controller circuit in TinkerCAD** and include a screenshot with the data sheet.
- Using the values provided for pin A0 and A1, **verify** what the voltage, current, and power measurements are on the **Serial Monitor** in a **TinkerCAD simulation** and record your results. Make sure to update the R_1 , R_2 , R_i , R_f values in the code.
- Physically build and test your circuit.** You can use a DC supply to replicate the A0 and A1 inputs for troubleshooting and testing your design. **Verify** what the voltage, current, and power measurements are on the **Serial Monitor** in your physical circuit and record your results.
- Create a **simulation of your Voltage Divider circuit (from Part 1) and Controller circuit together in TinkerCAD** and include a screenshot on the data sheet (you can use a DC voltage supply in TinkerCAD to replicate the solar panel).

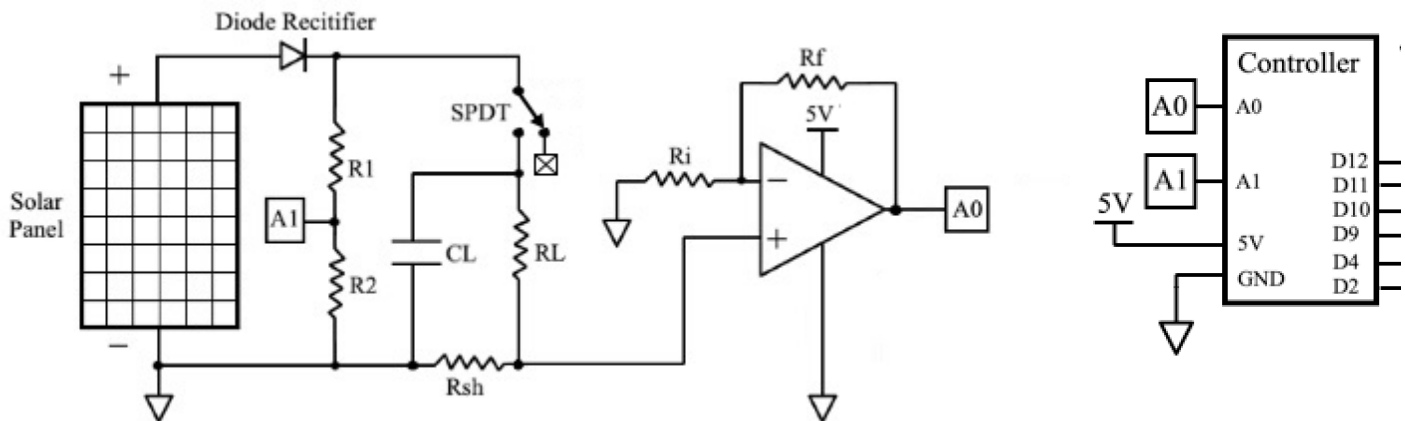


- The Controller is used to take the voltage value from pin A1 and then calculate what the actual solar panel voltage is and display it in the Serial Monitor. In a **TinkerCAD simulation**, set your solar panel voltage to the maximum expected voltage (from Part 1) – you can simulate the solar panel with a DC supply. **Record the Maximum Solar Panel Voltage and the voltage displayed on the Serial Monitor** – do they match as expected?
- Do your **simulated** results from your TinkerCAD circuit match your **theoretical** results? How much error are you seeing? What are some possible reasons for the difference?

j) **Physically build and test your Voltage divider and Controller circuit.** You can use a DC supply to replicate the solar panel for troubleshooting and testing your design but be sure to also take it outside to test with the solar panel.

k) **Do your measured results from your built circuit match your simulated results? How much error are you seeing? What are some possible reasons for the difference in the real and theoretical results?**

l) Create a **simulation of your Voltage Divider circuit (from Part 1) and Op-Amp circuit (from Part 2) and Controller circuit (Part 3) together in TinkerCAD** and include a screenshot on the data sheet (*you can use a DC voltage supply in TinkerCAD to replicate the solar panel*).



m) Set your solar panel voltage and current to the maximum expected voltage and current (from Parts 1 & 2). Record the measured **voltage**, **current**, and **power** from the simulated Serial Monitor on TinkerCAD. The 5V output from the Arduino can supply the op-amp.

n) Do your **simulated results** from your TinkerCAD circuit match your **theoretical results**? How much error are you seeing? What are some possible reasons for the difference?

o) **Physically build and test your full Volt/Amp/Wattmeter circuit.** You can use a DC supply to replicate the solar panel for troubleshooting and testing your design.

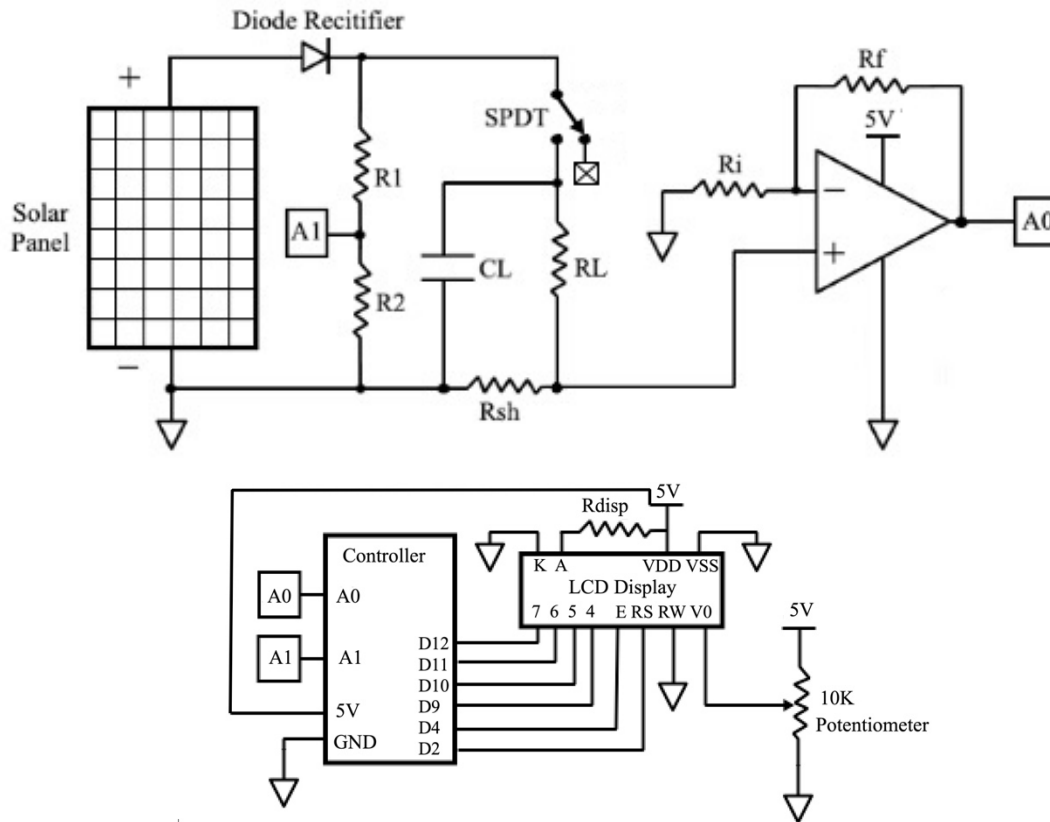
p) Do your **measured results** from your built circuit match your **simulated results**? How much error are you seeing? What are some possible reasons for the difference in the real and theoretical results?

q) What **adjustments** can you make to the code to **increase the accuracy** of the voltage, current, and power that display in the serial port?

Bonus Point Opportunity!

Considering the user of this power measuring device (solar installer or electronics enthusiast) and the application (portable, automated, innovative, user-friendly), add any additional automated features to enhance the experience of the user. Include your updated code and schematic attached to the data sheet. Bonus points will be given for successful changes in the code, and/or successful changes in the simulation, and/or successful changes in the physical prototype.

Step 4: Design the LCD Display circuit with your power measuring circuit.



Using your understanding of Arduino circuits and **LCD displays**, design and test the complete automated solar panel voltage, current, and power measurement circuit. Be sure to refer back to Lab 3 for details on the LCD display circuit set-up. For this design, set the display resistor (R_{disp}) to **220 Ω** .

For your LCD display, be sure to add to the initialization of your code:

```
#include <LiquidCrystal.h>
const int rs = 2, en = 4, d4 = 9, d5 = 10, d6 = 11, d7 = 12; //set digital output pins for LCD display
LiquidCrystal lcd(rs, en, d4, d5, d6, d7);
```

Add to the set-up of your code:

```
lcd.begin(16, 2); //set up LCD display

//write to LCD display to verify it's working
lcd.print("  Circuits I  ");
lcd.setCursor(0, 1);
lcd.print("  Wattmeter  ");
delay(2000);
lcd.clear();
```

Add to the main body of your code:

```
//Print values to LCD display
lcd.setCursor(0, 0);
lcd.print("V="); lcd.print(Voltage); //lcd.print("V");
lcd.print(" ");
lcd.print("P="); lcd.print(Power); lcd.print("mW ");
lcd.setCursor(0, 1);
lcd.print("I="); lcd.print(Current); lcd.print("mA");
lcd.print(" ");
delay(1000);
```

Data Sheet Part 4

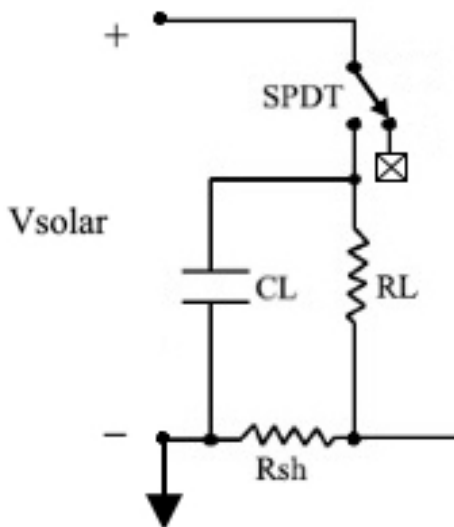
Answer all the questions in the data sheet.

- a) Create a **simulation of your Automated Solar Panel Power Meter circuit in TinkerCAD** and include a screenshot with the data sheet.
- b) **Test and troubleshoot** your simulation (you can use a DC supply to represent the solar panel) over the **given voltage supply values**. *Hint: Give the LCD a moment to adjust after changing the input voltage.*
- c) Did you have to make any component or code **changes** from part 3? If so, **what were the changes and why** did you make them?
- d) **Physically build and test your circuit.** You can use a DC supply to replicate the solar panel for troubleshooting and testing your design but be sure to also test with the actual solar panel in the full sunlight and shaded sunlight conditions (*for poor outside light conditions, use the provided indoor light for testing*). Note the conditions you measure.
- e) Do your **measured** results from your built circuit match your **simulated** results? How much error are you seeing? What are some possible reasons for the difference in the real and theoretical results?

Bonus Point Opportunity – use the LCD for additional purposes that somehow benefit the user. Consider the user of this power measuring device (solar installer or electronics enthusiast) and the application (portable, automated, innovative, user-friendly). Include your updated code and schematic attached to the data sheet. Bonus points will be given for successful changes in the code, and/or successful changes in the simulation, and/or successful changes in the physical prototype.

Step 5: Select a Load Capacitor to set the correct discharge time

When operating the Solar Panel Power Meter, the SPDT switch will connect the load to the solar panel. When the SPDT switch is open, the load capacitor (CL) will discharge.



Using **First-Order Transient Circuit analysis**, select a capacitor value (from your available parts) so the load discharges to approximately **1mV** in **10msec**. Assume a **6V, 5mA** solar panel supply and a load resistance (R_L) of **1 k Ω** . (You can assume 1mV to be a fully discharged output).

Data Sheet Part 5

Answer all the questions in the data sheet.

Using the First Order Differential method, First Order Step-by-Step method, or Laplace transforms, determine the size of the load capacitor to meet the design specifications noted above. Show your work on the data sheet.

Design Recipe: Selecting the Load Capacitor Using Discharge Time

Goal: Choose the load capacitor so that the output voltage decays over a specified time when the load draws current.

1. Determine the time constant

Start with the equation for a 1st order transient:

$$V(t) = V_o e^{-t/\tau}$$

Set the specifications:

$$t = \text{_____} s$$

$$V_o = \text{_____} V$$

$$V(t) = \text{_____} V$$

Solve for the time constant:

$$\tau = \frac{t}{-\ln\left(\frac{V(t)}{V_o}\right)}$$

2. Identify the Effective Load Resistance

Determine the resistance seen by the capacitor during discharge:

$$R_{eq} \approx R_L + R_{sh}$$

3. Use the RC Time Constant

For an RC discharge:

$$\tau = R_{eq} * C_L$$

Calculate the required C_L :

$$C_L = \frac{\tau}{R_{eq}}$$

Bonus Point Opportunity - Test your capacitor selection in Part 5 using your physical circuit from Part 4.

Verify your capacitor choice in your built design using an oscilloscope. Capture the discharge waveform and include a screenshot. Ask a TA for assistance if needed.

Prototype Demonstration

Answer all the questions in the data sheet.

Upload your Controller code to the Arduino and replace the USB input with the 9V battery, so you can take your design outside for additional testing. Record your results.

Demonstrate your working prototype to the Sparky Solar representative (TA) and be sure to get a signature on your data sheet.

Bonus Point Opportunity – *demonstrate any additional features you've added for possible extra credit.*

Lab Report

Along with the completed Data Sheet, include a final report that outlines a summary of the design project experience. The purpose of this report is to document your engineering decision-making, not to restate lab instructions. The report should be 4-10 pages long and should clearly demonstrate your contribution to the design work. This report will be submitted to a separate assignment in the course shell. Responses should reference specific design decisions, measurements, calculations, or troubleshooting steps from your project. This report is a technical document written in narrative form (paragraphs, not bullet points), past tense and should detail the following:

Design Intent and Stakeholders:

1. Who was this project designed for and what considerations were made for the user(s)?
2. If this design were scaled up for large solar panel systems, what environmental, technological, societal, and/or financial considerations would need to be made?

Technical Design and Decisions:

3. How does the final design work - give a detailed description with images?
4. What components did you specifically determine and explain your reasoning for your choices?
5. What trade-offs were made in this design?

Validation and Iteration:

6. What differences did you observe between your theoretical calculations or simulations and the behavior of the physical circuit? Describe at least one specific example and explain the cause of the discrepancy.
7. Describe at least one specific issue encountered during simulation, construction, or testing. Explain how you identified the cause, the steps you took to troubleshoot it, and the final correction.

Professional Growth:

8. Identify at least two technical or engineering process skills developed during this project (e.g., circuit debugging, measurement, simulation validation, design iteration). Explain how you applied each skill in this project.

Fill in your data sheet, print your simulation schematics, include your Lab report. One data sheet and lab report submitted for each team member. If a member is not participating in the work, do not share your data sheet information with them.

Project Deliverables

1. TA-signed data sheet (individual)
2. All simulation schematics
3. Demonstrated **final working prototype** (team)
4. Final written report (Individual)

The Lab Project Data Sheet and Report will serve as your lab final exam. These are individual submissions.

Lab CLOS:

- Understand and apply the application of linear electrical circuits.
- Application of resistive networks, capacitors (in a First Order Transient circuit), and Op-Amps.
- Use SPICE and/or TinkerCAD to analyze a circuit and calculate expected values.
- Proficient in circuit simulation and measurement of electrical systems using a multimeter and an oscilloscope.
- Analysis and design using Ohm's Law, Kirchhoff's Laws and the Power Law.
- Understands the difference between calculated and measured results.
- Design of resistive networks and an Op-Amp closed loop circuit.
- Explores multiple solution paths.
- Applies technical skills/knowledge to the development of a technology/product.
- Focuses on understanding the value proposition of a discovery.
- Engages in actions with the understanding that they have potential to lead to both gains or losses.
- Persuades why a discovery adds value from multiple perspectives (technological, societal, financial, environmental, etc.).
- Identifies and works with individuals with complementary skill sets, expertise, etc.
- Integrates/synthesizes different kinds of knowledge.